f analyse the results obtained from the following experiments:

  i measuring the pH of a variety of substances, eg equimolar solutions of strong and weak acids, strong and weak bases and salts

  ii comparing the pH of a strong acid and a weak acid after dilution 10, 100 and 1000 times

g analyse and evaluate the results obtained from experiments to determine K_a for a weak acid by measuring the pH of a solution containing a known mass of acid, and discuss the assumptions made in this calculation

h calculate the pH of a solution of a weak acid based on data for concentration and K_a, and discuss the assumptions made in this calculation

i measure the pH change during titrations and draw titration curves using different combinations of strong and weak monobasic acids and bases

j use data about indicators, together with titration curves, to select a suitable indicator and the use of titrations in analysis

k explain the action of buffer solutions and carry out calculations on the pH of buffer solutions, eg making buffer solutions and comparing the effect of adding acid or alkali on the pH of the buffer

l use titration curves to show the buffer action and to determine K_a from the pH at the point where half the acid is neutralised

m explain the importance of buffer solutions in biological environments, eg buffers in cells and in blood (H_2CO_3/HCO_3^-) and in foods to prevent deterioration due to pH change (caused by bacterial or fungal activity).

4.8 Further organic chemistry

Related topics in *Unit 5* will assume knowledge of this material.

During this topic there will be the opportunity to carry out an internal assessment activity. Please see *Appendix 9* for more details.

Students will be assessed on their ability to:

1 Chirality

a recall the meaning of structural and E-Z isomerism (geometric/cis-trans isomerism)

b demonstrate an understanding of the existence of optical isomerism resulting from chiral centre(s) in a molecule with asymmetric carbon atom(s) and understand optical isomers as object and non-superimposable mirror images

c recall optical activity as the ability of a single optical isomer to rotate the plane of polarization of plane-polarized monochromatic light in molecules containing a single chiral centre and understand the nature of a racemic mixture

d use data on optical activity of reactants and products as evidence for proposed mechanisms, as in S_N1 and S_N2 and addition to carbonyl compounds.

2 Carbonyl compounds

a give examples of molecules that contain the aldehyde or ketone functional group

b explain the physical properties of aldehydes and ketones relating this to the lack of hydrogen bonding between molecules and their solubility in water in terms of hydrogen bonding with the water

c describe and carry out, where appropriate, the reactions of carbonyl compounds. This will be limited to:

- i oxidation with Fehling's or Benedict's solution, Tollens' reagent and acidified dichromate(VI) ions
- ii reduction with lithium tetrahydridoaluminate (lithium aluminium hydride) in dry ether
- iii nucleophilic addition of HCN in the presence of KCN, using curly arrows, relevant lone pairs, dipoles and evidence of optical activity to show the mechanism
- iv the reaction with 2,4-dinitrophenylhydrazine and its use to detect the presence of a carbonyl group and to identify a carbonyl compound given data of the melting temperatures of derivatives
- v iodine in the presence of alkali.